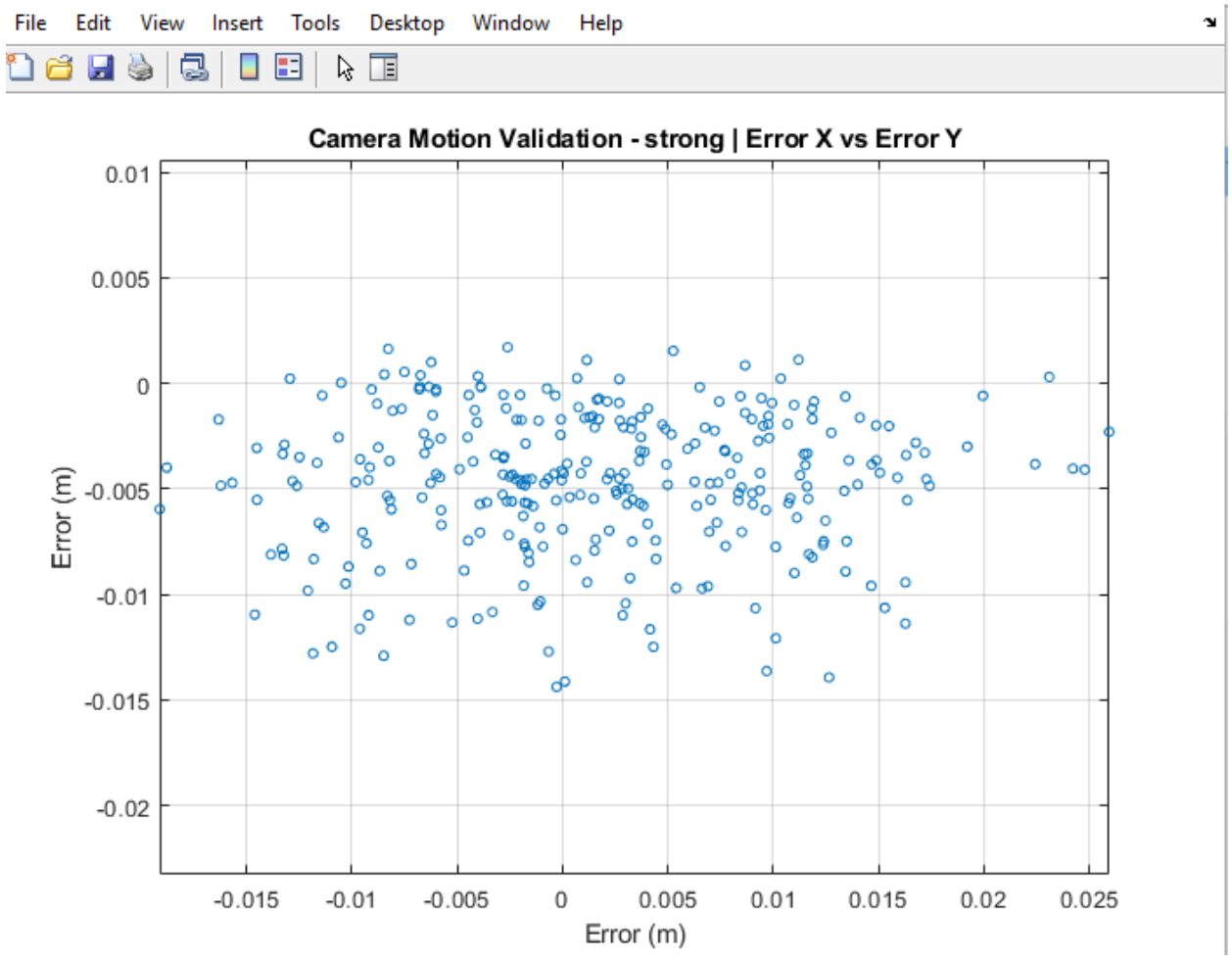
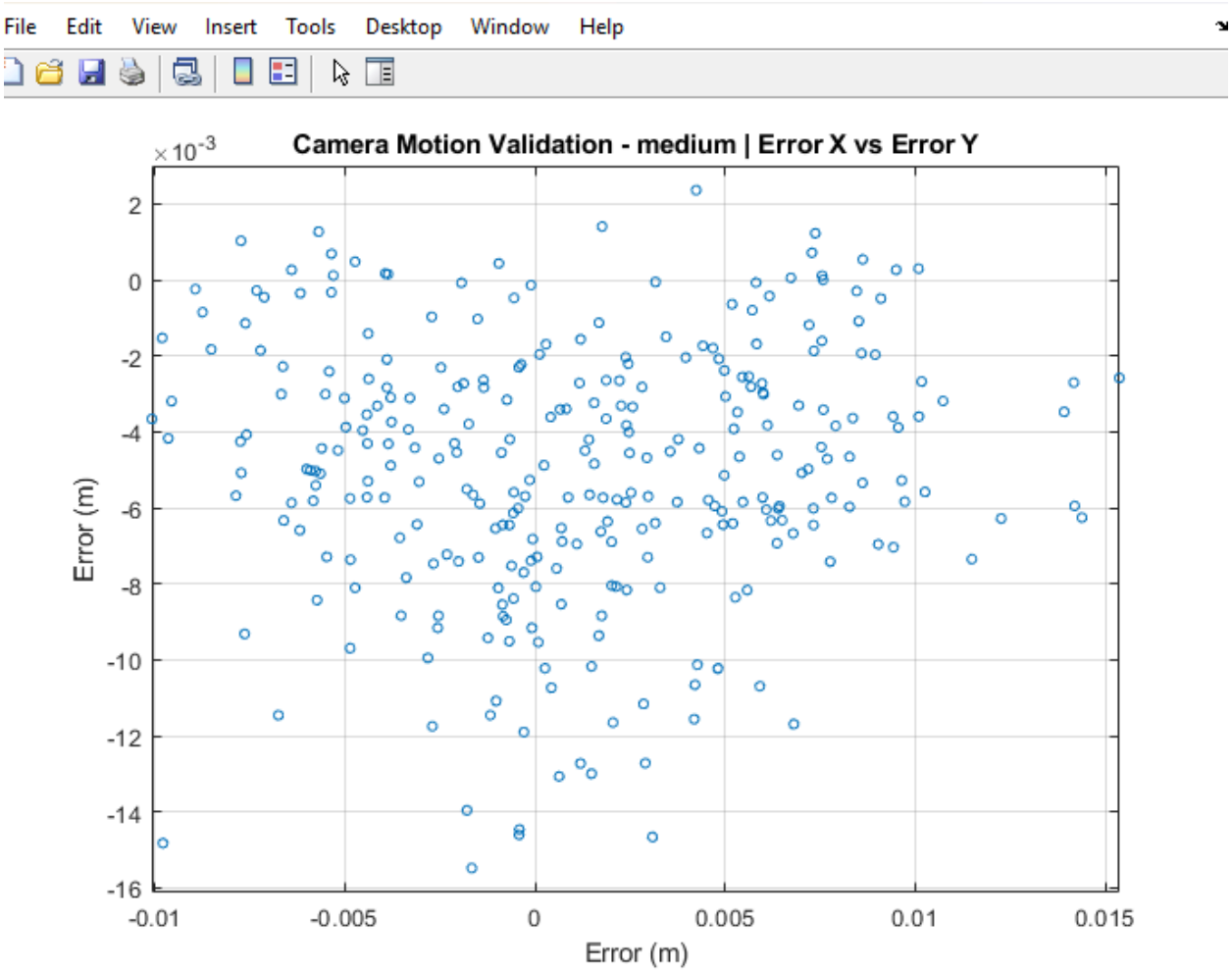
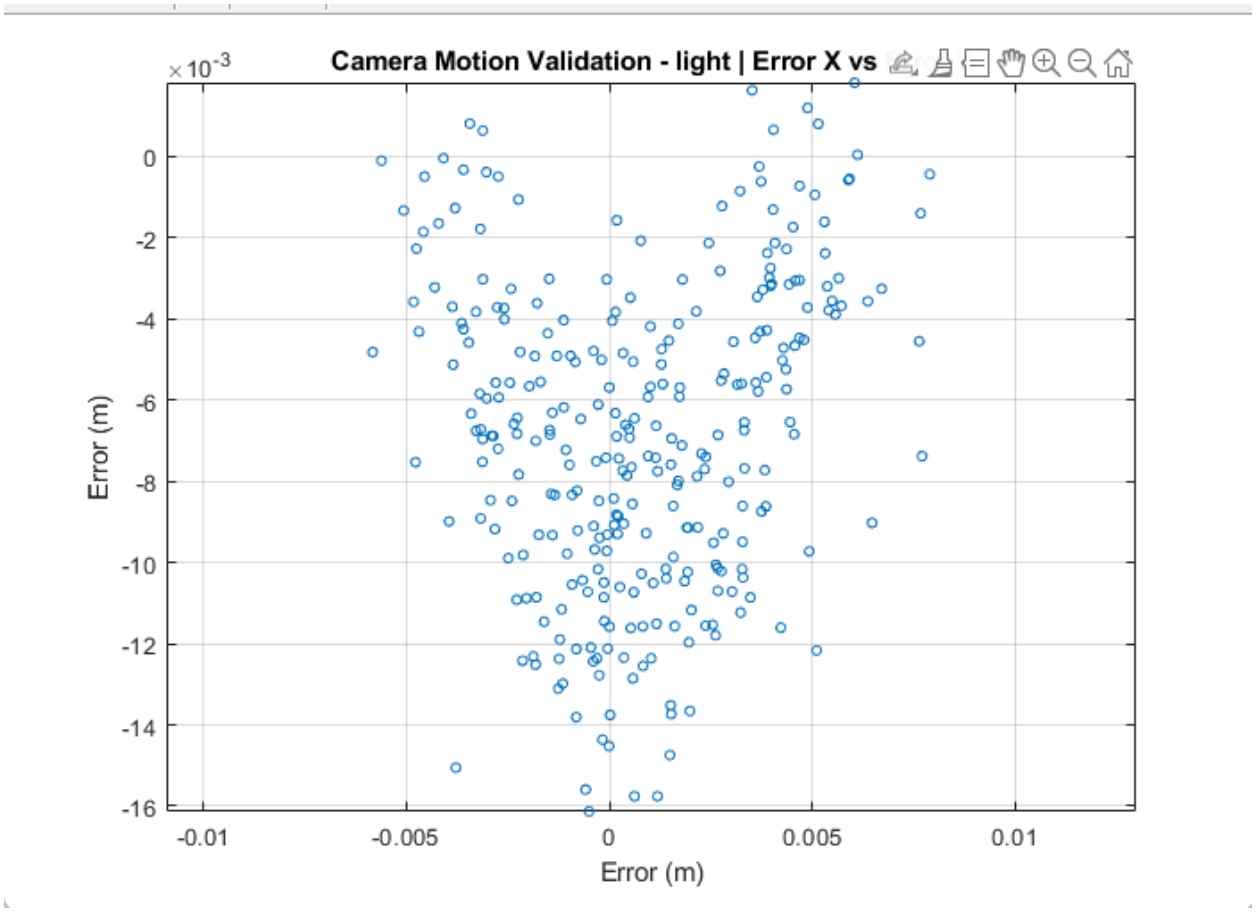
Wind Plots:


File Edit View Insert Tools Desktop Window Help
Camera Motion Validation - strong | Error X vs Error Y
0.01
0.005
0
-0.005
-0.01
-0.015
-0.02
Error (m)
-0.015
-0.01
-0.005
0
0.005
0.01
0.015
0.02
0.025
Error (m)


File Edit View Insert Tools Desktop Window Help
Camera Motion Validation - medium | Error X vs Error Y
×10^-3
2
0
-2
-4
-6
-8
-10
-12
-14
-16
Error (m)
-0.01
-0.005
0
0.005
0.01
0.015
Error (m)



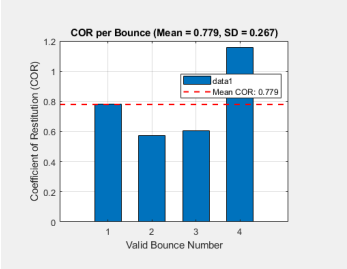

```
=== STRONG WIND SUMMARY ===
       label: "Camera Motion Validation - strong"
     nTrials: 300
      nValid: 300
   validFrac: 1
       meanE: 0.0105
      rmseE: 0.0118
        maxE: 0.0260
        p95E: 0.0200
        p99E: 0.0241
      meanDx: 0.0016
      meanDy: -0.0047
      meanDz: 0.0022
    maxDxAbs: 0.0259
    maxDyAbs: 0.0144
    maxDzAbs: 0.0147
    worstRow: [1×36 table]

=== ALL WIND LEVELS SUMMARY ===
```

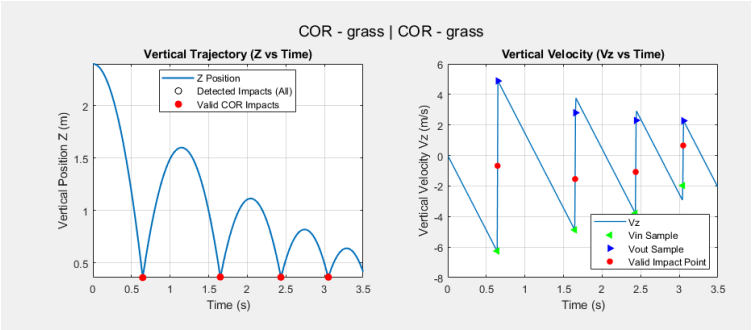
```
   WindLevel    ValidFrac     MeanE       RMSE        P95E       MaxE
   _________    _________    _________    _________    _______    _______

   {'light' }        1      0.010102     0.011209    0.018793    0.022611
   {'medium'}    0.99333     0.0087848    0.0096383    0.017267    0.024179
   {'strong'}        1      0.010526     0.011767    0.019977    0.026039

>>



Surface comparison:
>> [summaryTbl, allResults] = run_cor_surface_comparison( ...
   'GroundZ', 0.35951, ...
   'X0', 0, ...
   'Y0', -7.5, ...
   'Z0', 2.4, ...
   'Duration', 3.5, ...
   'SampleDt', 0.005, ...
   'VelocityPointsOffset', 1, ...
   'SmoothWindow', 1, ...
   'PlotResults', true);
Detected bounce indices:
  130   298   407   478   524   554   573   586

Detected bounce times:
   0.6450    1.4850    2.0300    2.3850    2.6150    2.7650    2.8600    2.9250

Warning: Ignoring extra legend entries.
> In legend>process_inputs (line 592)
In legend>make_legend (line 319)
In legend (line 263)
In run_cor_bounce_test (line 190)
In run_cor_surface_comparison (line 38)

Processing bounces for COR:
--------------------------------------------------
Bounce | Impact Time (s) | Vin (m/s) | Vout (m/s) | COR
--------------------------------------------------
1      | 0.645           | -6.260    | 4.064      | 0.649
2      | 1.485           | -3.033    | 2.609      | 0.860
3      | 2.030           | -1.964    | 1.674      | 0.852
4      | 2.385           | -1.059    | 1.058      | 0.999
5      | 2.615           | -0.726    | 0.665      | 0.917
```

```
6      | 2.765          | -0.425     | 0.405       | 0.953
7      | 2.860          | -0.425     | 0.261       | 0.613
8      | 2.925          | -0.186     | 0.135       | 0.724
--------------------------------------------------

Overall COR results:
  Number of valid bounces analyzed: 8
  Mean Coefficient of Restitution (COR): 0.8210
  Standard Deviation of COR:          0.1429
Detected bounce indices:
   130   316   450   546   615   665

Detected bounce times:
    0.6450    1.5750    2.2450    2.7250    3.0700    3.3200

Warning: Ignoring extra legend entries.
> In legend>process_inputs (line 592)
In legend>make_legend (line 319)
In legend (line 263)
In run_cor_bounce_test (line 190)
In run_cor_surface_comparison (line 38)

Processing bounces for COR:
--------------------------------------------------
Bounce | Impact Time (s) | Vin (m/s) | Vout (m/s) | COR
--------------------------------------------------
1      | 0.645          | -6.259     | 4.506       | 0.720
2      | 1.575          | -3.544     | 3.219       | 0.908
4      | 2.725          | -1.897     | 1.641       | 0.865
5      | 3.070          | -1.645     | 1.002       | 0.609
6      | 3.320          | -1.173     | 0.790       | 0.673
--------------------------------------------------

Overall COR results:
  Number of valid bounces analyzed: 5
  Mean Coefficient of Restitution (COR): 0.7551
  Standard Deviation of COR:          0.1273
Detected bounce indices:
   130   331   488   611

Detected bounce times:
    0.6450    1.6500    2.4350    3.0500

Warning: Ignoring extra legend entries.
```

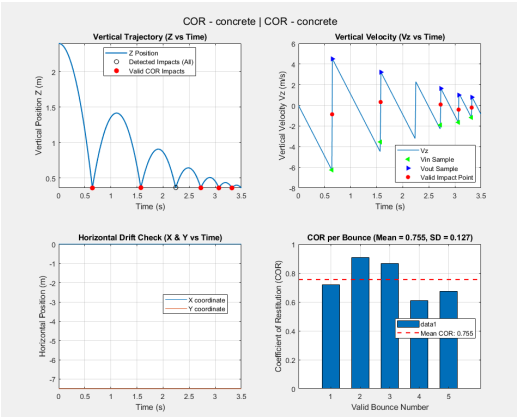
```
> In legend>process_inputs (line 592)
In legend>make_legend (line 319)
In legend (line 263)
In run_cor_bounce_test (line 190)
In run_cor_surface_comparison (line 38)

Processing bounces for COR:
--------------------------------------------------
Bounce | Impact Time (s) | Vin (m/s) | Vout (m/s) | COR
--------------------------------------------------
1      | 0.645           | -6.258    | 4.886      | 0.781
2      | 1.650           | -4.875    | 2.798      | 0.574
3      | 2.435           | -3.791    | 2.299      | 0.607
4      | 3.050           | -1.969    | 2.275      | 1.155
--------------------------------------------------

Overall COR results:
  Number of valid bounces analyzed: 4
  Mean Coefficient of Restitution (COR): 0.7790
  Standard Deviation of COR:          0.2666
=== SURFACE COR COMPARISON ===
     Surface       ModeledEApprox    MeasuredMeanCOR    MeasuredStdCOR
NumValidBounces
   _____    _______________    _________________    ________________
_________________

   {'clay'    }        0.65            0.82097          0.14291            8
   {'concrete'}        0.72            0.75511          0.12733            5
   {'grass'   }        0.78            0.77904          0.2666             4

>>
```

Grass:

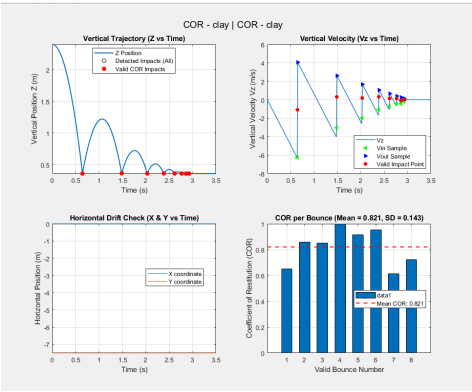


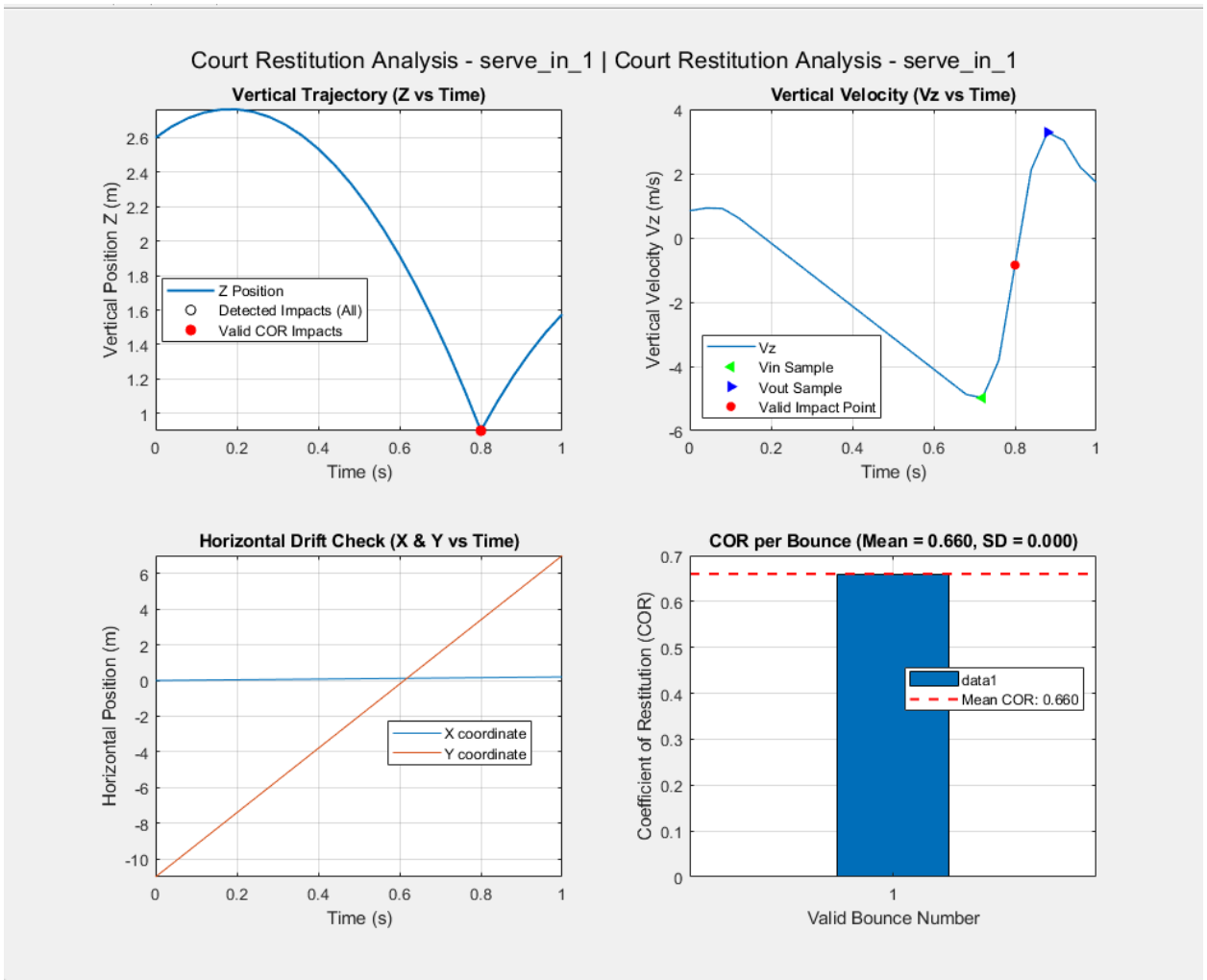


Concrete:



Clay:





IF a user wanted to use their own court data:

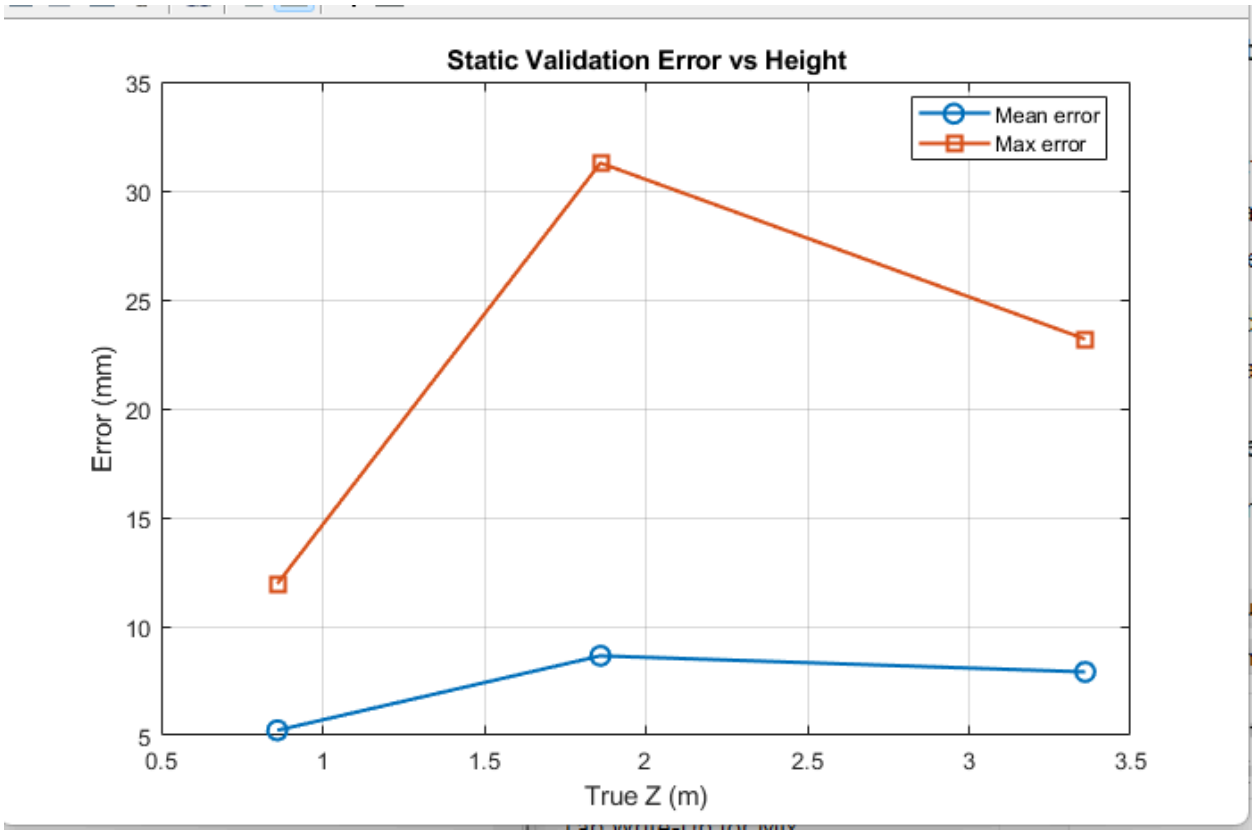
A user could collect court data and then use **only** `analyze_court_restitution.m` to calculate COR from real measurements. That function already supports:

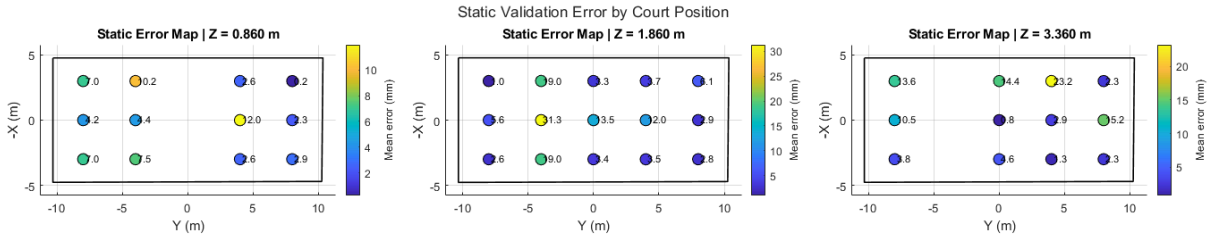
CSV input

MATLAB

```
outCOR = analyze_court_restitution('my_real_court_bounce_data.csv', ...
    'GroundZ', 0.35951, ...
    'PlotResults', true, ...
    'Label', 'Real Court COR');
```

Overall Accuracy:





=== STATIC VALIDATION ===

Xtrue	Ytrue	Ztrue	Xmeas	Ymeas	Zmeas	dX	dY	dZ	e	pairIdx	repeatIdx
-3	-8	0.85951	-2.9987	-8.0035	0.86543	0.0012645	-0.0035029	0.005918	0.0069922	1	1
-3	-8	0.85951	-2.9987	-8.0035	0.86543	0.0012645	-0.0035029	0.005918	0.0069922	1	2
-3	-8	0.85951	-2.9987	-8.0035	0.86543	0.0012645	-0.0035029	0.005918	0.0069922	1	3
0	-8	0.85951	-1.1205e-15	-7.9981	0.85576	-1.1205e-15	0.0019246	-0.0037477	0.004213	1	1
0	-8	0.85951	-1.1205e-15	-7.9981	0.85576	-1.1205e-15	0.0019246	-0.0037477	0.004213	1	2
0	-8	0.85951	-1.1205e-15	-7.9981	0.85576	-1.1205e-15	0.0019246	-0.0037477	0.004213	1	3
3	-8	0.85951	2.9987	-8.0035	0.86543	-0.0012645	-0.0035029	0.005918	0.0069922	1	1
3	-8	0.85951	2.9987	-8.0035	0.86543	-0.0012645	-0.0035029	0.005918	0.0069922	1	2
3	-8	0.85951	2.9987	-8.0035	0.86543	-0.0012645	-0.0035029	0.005918	0.0069922	1	3
-3	-4	0.85951	-2.9982	-4.007	0.86667	0.0017934	-0.0070067	0.0071623	0.010179	1	1
-3	-4	0.85951	-2.9982	-4.007	0.86667	0.0017934	-0.0070067	0.0071623	0.010179	1	2
-3	-4	0.85951	-2.9982	-4.007	0.86667	0.0017934	-0.0070067	0.0071623	0.010179	1	3
0	-4	0.85951	-0.0034082	-4.0019	0.86155	-0.0034082	-0.0019259	0.0020367	0.0044128	1	1

0	-4	0.85951	-0.0034082	-4.0019	0.86155	-0.0034082	-0.0019259	0.0020367	0.0044128	1	2
0	-4	0.85951	-0.0034082	-4.0019	0.86155	-0.0034082	-0.0019259	0.0020367	0.0044128	1	3
3	-4	0.85951	2.9989	-4.005	0.86493	-0.0010507	-0.0050299	0.0054163	0.007466	1	1
3	-4	0.85951	2.9989	-4.005	0.86493	-0.0010507	-0.0050299	0.0054163	0.007466	1	2
3	-4	0.85951	2.9989	-4.005	0.86493	-0.0010507	-0.0050299	0.0054163	0.007466	1	3
-3	0	0.85951	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	1
-3	0	0.85951	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	2
-3	0	0.85951	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	3
0	0	0.85951	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	1
0	0	0.85951	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	2
0	0	0.85951	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	3
3	0	0.85951	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	1
3	0	0.85951	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	2
3	0	0.85951	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	3
-3	4	0.85951	-3.0004	3.9984	0.86145	-0.00044208	-0.0016452	0.0019365	0.0025792	2	1
-3	4	0.85951	-3.0004	3.9984	0.86145	-0.00044208	-0.0016452	0.0019365	0.0025792	2	2
-3	4	0.85951	-3.0004	3.9984	0.86145	-0.00044208	-0.0016452	0.0019365	0.0025792	2	3
0	4	0.85951	-0.0024941	3.9935	0.86919	-0.0024941	-0.0065489	0.0096848	0.011954	2	1
0	4	0.85951	-0.0024941	3.9935	0.86919	-0.0024941	-0.0065489	0.0096848	0.011954	2	2
0	4	0.85951	-0.0024941	3.9935	0.86919	-0.0024941	-0.0065489	0.0096848	0.011954	2	3
3	4	0.85951	3.0004	3.9984	0.86145	0.00044208	-0.0016452	0.0019365	0.0025792	2	1
3	4	0.85951	3.0004	3.9984	0.86145	0.00044208	-0.0016452	0.0019365	0.0025792	2	2

3 4 0.85951 3.0004 3.9984 0.86145 0.00044208 -0.0016452
0.0019365 0.0025792 2 3
-3 8 0.85951 -2.9998 8.0001 0.85939 0.00020242 6.6352e-05
-0.00011983 0.00024441 2 1
-3 8 0.85951 -2.9998 8.0001 0.85939 0.00020242 6.6352e-05
-0.00011983 0.00024441 2 2
-3 8 0.85951 -2.9998 8.0001 0.85939 0.00020242 6.6352e-05
-0.00011983 0.00024441 2 3
0 8 0.85951 -1.0496e-15 8.0018 0.85806 -1.0496e-15 0.0017726
-0.0014504 0.0022904 2 1
0 8 0.85951 -1.0496e-15 8.0018 0.85806 -1.0496e-15 0.0017726
-0.0014504 0.0022904 2 2
0 8 0.85951 -1.0496e-15 8.0018 0.85806 -1.0496e-15 0.0017726
-0.0014504 0.0022904 2 3
3 8 0.85951 2.9995 7.9979 0.86138 -0.00048237 -0.0021051
0.0018676 0.0028551 2 1
3 8 0.85951 2.9995 7.9979 0.86138 -0.00048237 -0.0021051
0.0018676 0.0028551 2 2
3 8 0.85951 2.9995 7.9979 0.86138 -0.00048237 -0.0021051
0.0018676 0.0028551 2 3
-3 -8 1.8595 -3.0008 -8 1.8588 -0.00075452 2.0699e-06
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-0.00072198 0.0010443 1 2
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-0.012097 0.018998 1 3

0 -4 1.8595 0.00099043 -4.0218 1.8819 0.00099043 -0.021774
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-3 0 1.8595 -3.0015 0.0029102 1.8589 -0.0014737 0.0029102
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0.0098275 0.011986 2 2
0 4 1.8595 0.0011706 3.9932 1.8693 0.0011706 -0.0067618
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0.0028954 0.003517 2 2
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0.0028954 0.003517 2 3
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-0.0047824 0.0061425 2 1
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-0.0047824 0.0061425 2 2
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0 8 1.8595 -1.4508e-15 7.9973 1.8606 -1.4508e-15 -0.0026658
0.0011301 0.0028954 2 1
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0.0011301 0.0028954 2 2
0 8 1.8595 -1.4508e-15 7.9973 1.8606 -1.4508e-15 -0.0026658
0.0011301 0.0028954 2 3
3 8 1.8595 3.0005 8.0015 1.8572 0.00053485 0.0015358
-0.0022662 0.0027893 2 1
3 8 1.8595 3.0005 8.0015 1.8572 0.00053485 0.0015358
-0.0022662 0.0027893 2 2
3 8 1.8595 3.0005 8.0015 1.8572 0.00053485 0.0015358
-0.0022662 0.0027893 2 3
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0.0097872 0.013639 1 1
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0.0097872 0.013639 1 2
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0.0082998 0.010538 1 3
3 -8 3.3595 3.0017 -7.9979 3.3569 0.001716 0.0021024
-0.0025942 0.0037543 1 1
3 -8 3.3595 3.0017 -7.9979 3.3569 0.001716 0.0021024
-0.0025942 0.0037543 1 2
3 -8 3.3595 3.0017 -7.9979 3.3569 0.001716 0.0021024
-0.0025942 0.0037543 1 3
-3 -4 3.3595 NaN NaN NaN NaN NaN NaN
NaN NaN 1
-3 -4 3.3595 NaN NaN NaN NaN NaN NaN
NaN NaN 2

-3 -4 3.3595 NaN NaN NaN NaN NaN NaN
NaN NaN 3
0 -4 3.3595 NaN NaN NaN NaN NaN NaN
NaN NaN 1
0 -4 3.3595 NaN NaN NaN NaN NaN NaN
NaN NaN 2
0 -4 3.3595 NaN NaN NaN NaN NaN NaN
NaN NaN 3
3 -4 3.3595 NaN NaN NaN NaN NaN NaN
NaN NaN 1
3 -4 3.3595 NaN NaN NaN NaN NaN NaN
NaN NaN 2
3 -4 3.3595 NaN NaN NaN NaN NaN NaN
NaN NaN 3
-3 0 3.3595 -2.9947 -0.0052828 3.3719 0.005282 -0.0052828
0.012367 0.014448 2 1
-3 0 3.3595 -2.9947 -0.0052828 3.3719 0.005282 -0.0052828
0.012367 0.014448 2 2
-3 0 3.3595 -2.9947 -0.0052828 3.3719 0.005282 -0.0052828
0.012367 0.014448 2 3
0 0 3.3595 0.00012806 -0.00071444 3.3591 0.00012806 -0.00071444
-0.00037929 0.00081895 2 1
0 0 3.3595 0.00012806 -0.00071444 3.3591 0.00012806 -0.00071444
-0.00037929 0.00081895 2 2
0 0 3.3595 0.00012806 -0.00071444 3.3591 0.00012806 -0.00071444
-0.00037929 0.00081895 2 3
3 0 3.3595 2.9994 -0.0022791 3.3634 -0.00062245 -0.0022791
0.0038932 0.004554 2 1
3 0 3.3595 2.9994 -0.0022791 3.3634 -0.00062245 -0.0022791
0.0038932 0.004554 2 2
3 0 3.3595 2.9994 -0.0022791 3.3634 -0.00062245 -0.0022791
0.0038932 0.004554 2 3
-3 4 3.3595 -2.9921 3.9864 3.3765 0.0079252 -0.013591
0.017035 0.023189 2 1
-3 4 3.3595 -2.9921 3.9864 3.3765 0.0079252 -0.013591
0.017035 0.023189 2 2
-3 4 3.3595 -2.9921 3.9864 3.3765 0.0079252 -0.013591
0.017035 0.023189 2 3
0 4 3.3595 0.0002241 3.9983 3.3619 0.0002241 -0.0017191
0.0023427 0.0029144 2 1
0 4 3.3595 0.0002241 3.9983 3.3619 0.0002241 -0.0017191
0.0023427 0.0029144 2 2
0 4 3.3595 0.0002241 3.9983 3.3619 0.0002241 -0.0017191
0.0023427 0.0029144 2 3

3 4 3.3595 3.0005 3.9998 3.3607 0.00047603 -0.00016034 0.0011912 0.0012927 2 1
3 4 3.3595 3.0005 3.9998 3.3607 0.00047603 -0.00016034 0.0011912 0.0012927 2 2
3 4 3.3595 3.0005 3.9998 3.3607 0.00047603 -0.00016034 0.0011912 0.0012927 2 3
-3 8 3.3595 -3.0016 8.0009 3.3609 -0.0016006 0.00089641 0.0014218 0.002321 2 1
-3 8 3.3595 -3.0016 8.0009 3.3609 -0.0016006 0.00089641 0.0014218 0.002321 2 2
-3 8 3.3595 -3.0016 8.0009 3.3609 -0.0016006 0.00089641 0.0014218 0.002321 2 3
0 8 3.3595 -1.3484e-15 7.9893 3.3703 -1.3484e-15 -0.010684 0.010745 0.015152 2 1
0 8 3.3595 -1.3484e-15 7.9893 3.3703 -1.3484e-15 -0.010684 0.010745 0.015152 2 2
0 8 3.3595 -1.3484e-15 7.9893 3.3703 -1.3484e-15 -0.010684 0.010745 0.015152 2 3
3 8 3.3595 3.0016 8.0009 3.3609 0.0016006 0.00089641 0.0014218 0.002321 2 1
3 8 3.3595 3.0016 8.0009 3.3609 0.0016006 0.00089641 0.0014218 0.002321 2 2
3 8 3.3595 3.0016 8.0009 3.3609 0.0016006 0.00089641 0.0014218 0.002321 2 3

=== SHOT VALIDATION ===

preset	type	n	meanDx	meanDy	meanDz	meanE	maxE	rmseE	bounceE
"serve_default"	"serve"	8	1.3095e-05	-0.0018891	0.0028604	0.0037875	0.0076668	0.0044489	0.0021044
"serve_t"	"serve"	8	0.00096076	-0.0070292	0.0083198	0.011354	0.029327	0.01492	0.0020608
"serve_wide"	"serve"	9	-0.00020914	0.00073491	0.00046394	0.0053724	0.019664	0.0078967	0.0017737
"serve_body"	"serve"	8	-0.00064402	0.002445	-0.00064376	0.0059998	0.02029	0.0084583	0.0063952
"volley_default"	"volley"	5	0.00074243	-0.0031603	0.0069692	0.0094934	0.021234	0.011434	0.0057975
"volley_short"	"volley"	5	0.0012548	0.0014573	-0.005777	0.010724	0.016683	0.011295	0.0097759

```
   "volley_deep"      "volley"    5     -0.0010012     -0.004152      0.0030387     0.010594
0.036338     0.016815     0.036338
   "volley_cross"     "volley"    5    -5.3478e-05    -0.0033317      0.0070591    0.0089502
0.023267     0.012413    0.0040846

=== STATIC SUMMARY ===
     meanDx       meanDy       meanDz       meanE       maxE       rmseE      pair1Count
pair2Count
   __________   __________   _________   __________   _________   _________
__________   __________

   0.00040458    -0.0018532    0.003178    0.0073703    0.031272    0.010076       45          72

=== SHOT SUMMARY ===
   serveMeanE    serveMaxE    serveBounce    volleyMeanE    volleyMaxE    volleyBounce
   __________    _________    ___________    ___________    __________    ____________

   0.0066284     0.029327     0.0030835      0.0099403      0.036338      0.013999

>>
```

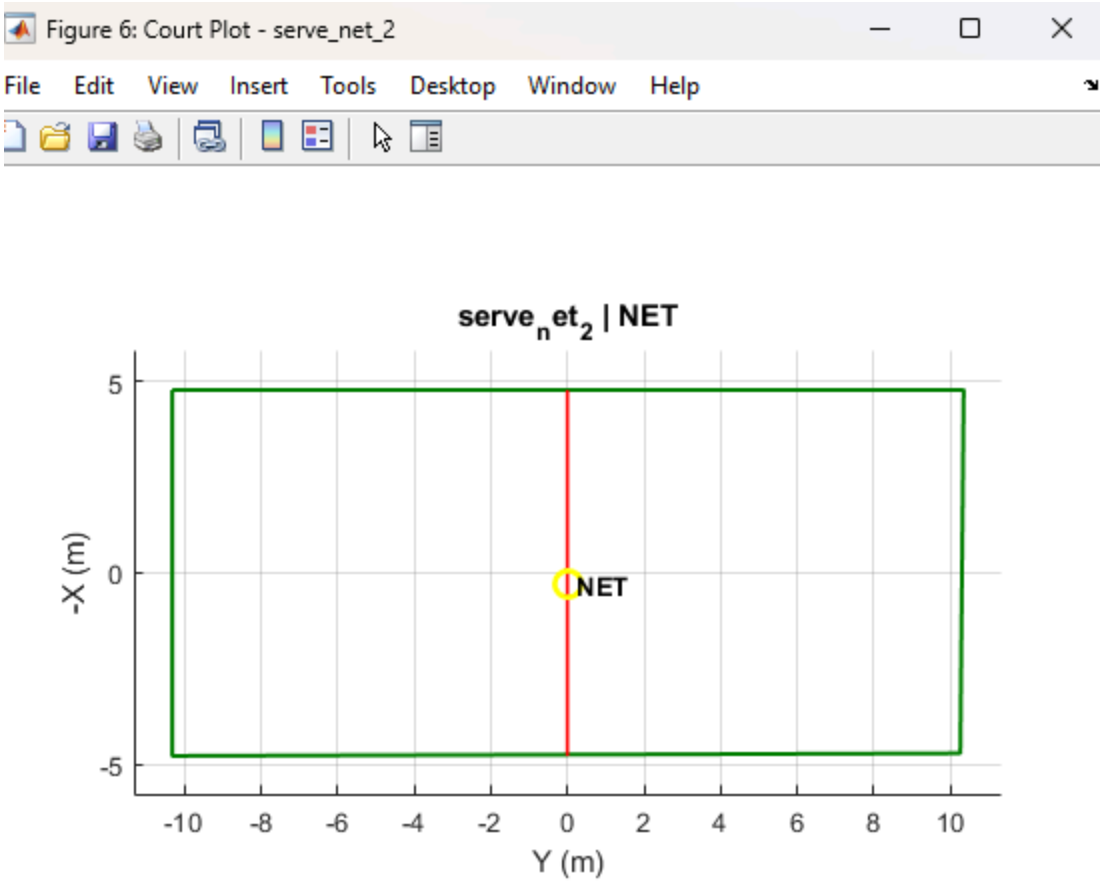
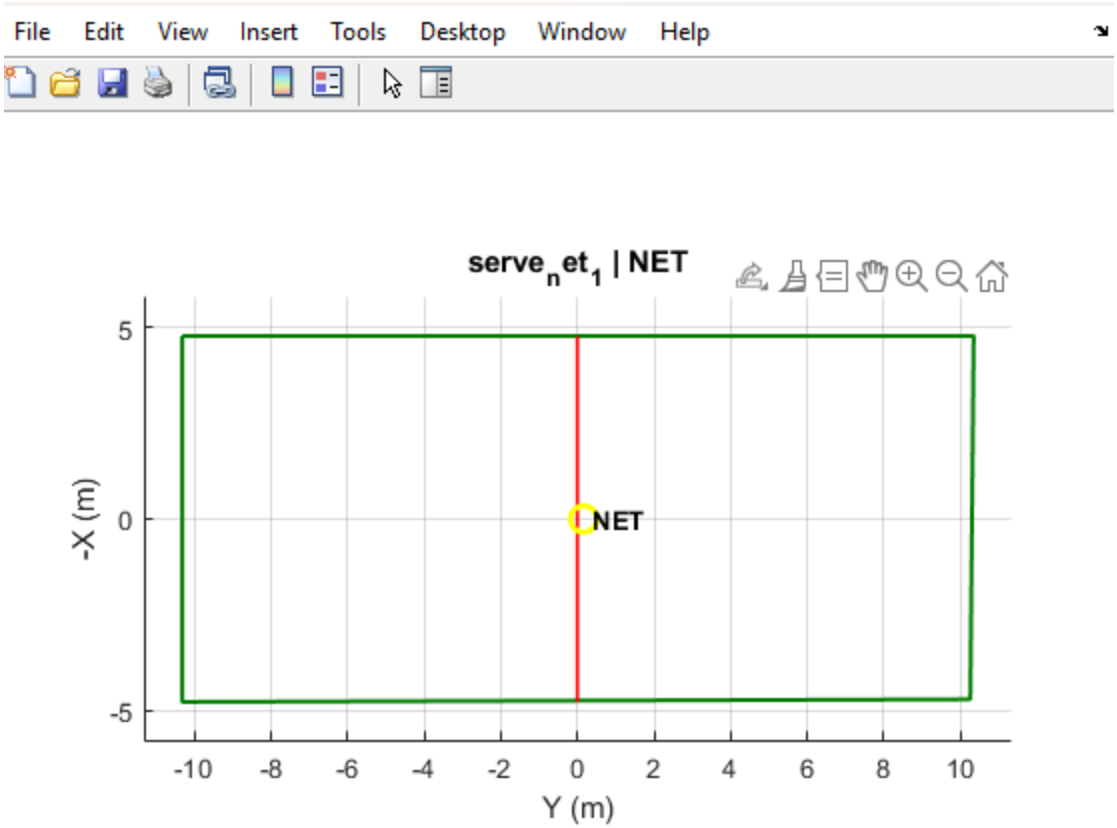
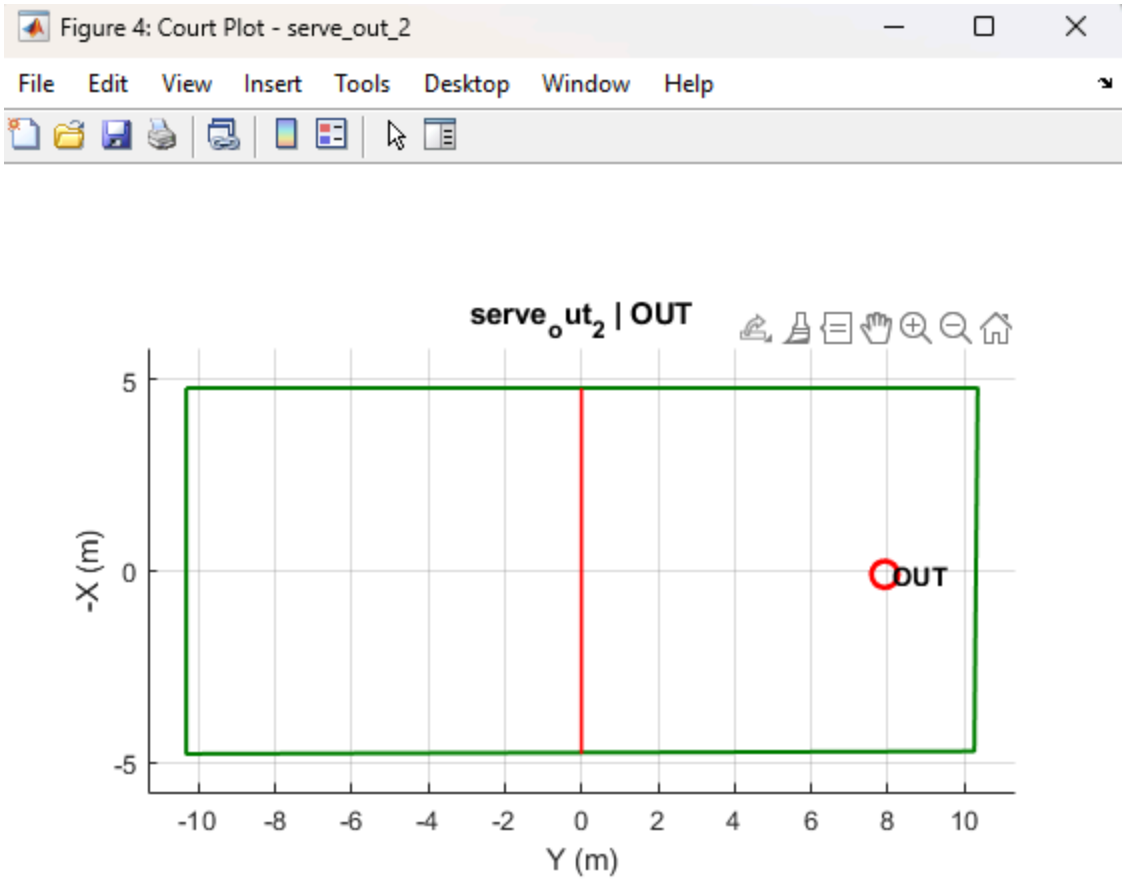
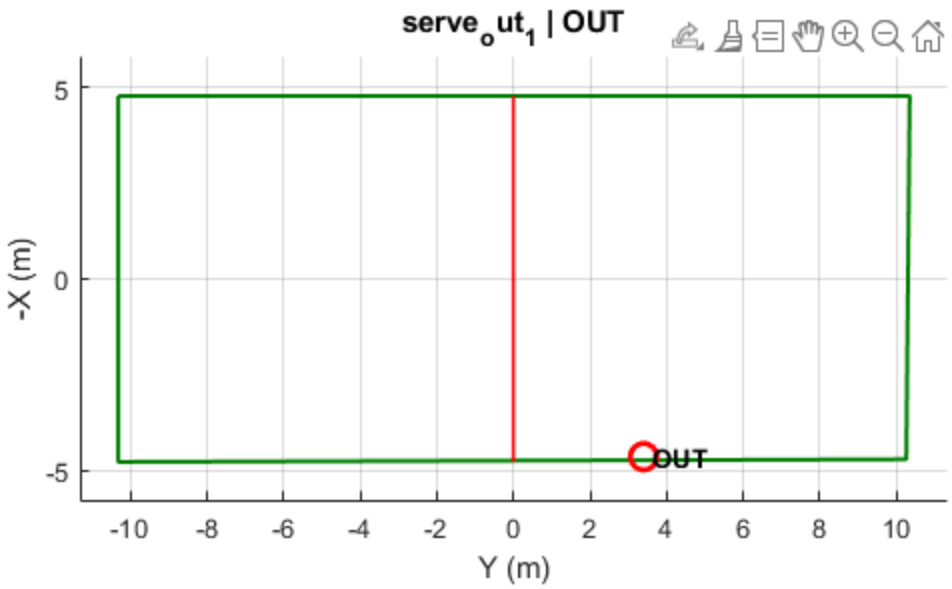


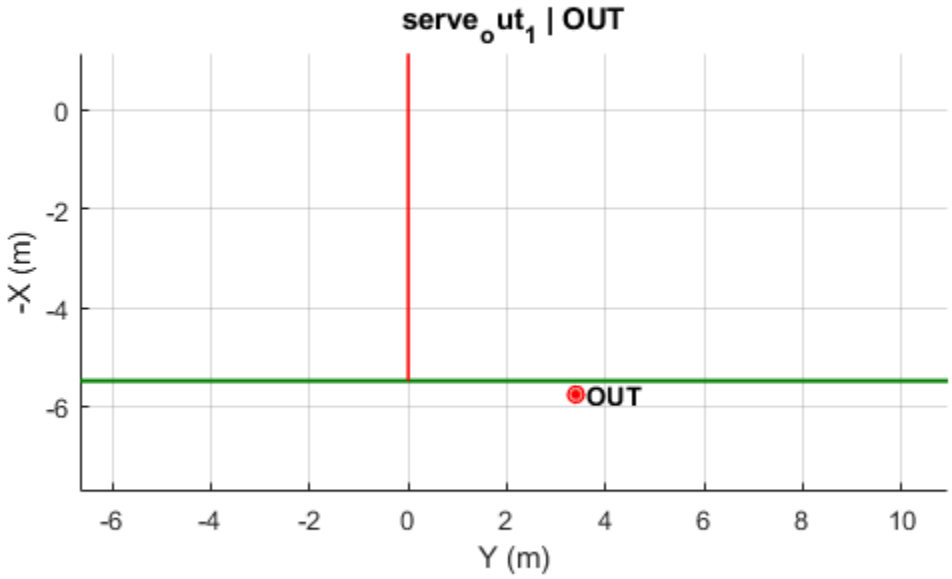
Figure 6: Court Plot - serve_net_2
File Edit View Insert Tools Desktop Window Help
serve_net_2 | NET
NET
-X (m)
Y (m)
5
0
-5
-10 -8 -6 -4 -2 0 2 4 6 8 10

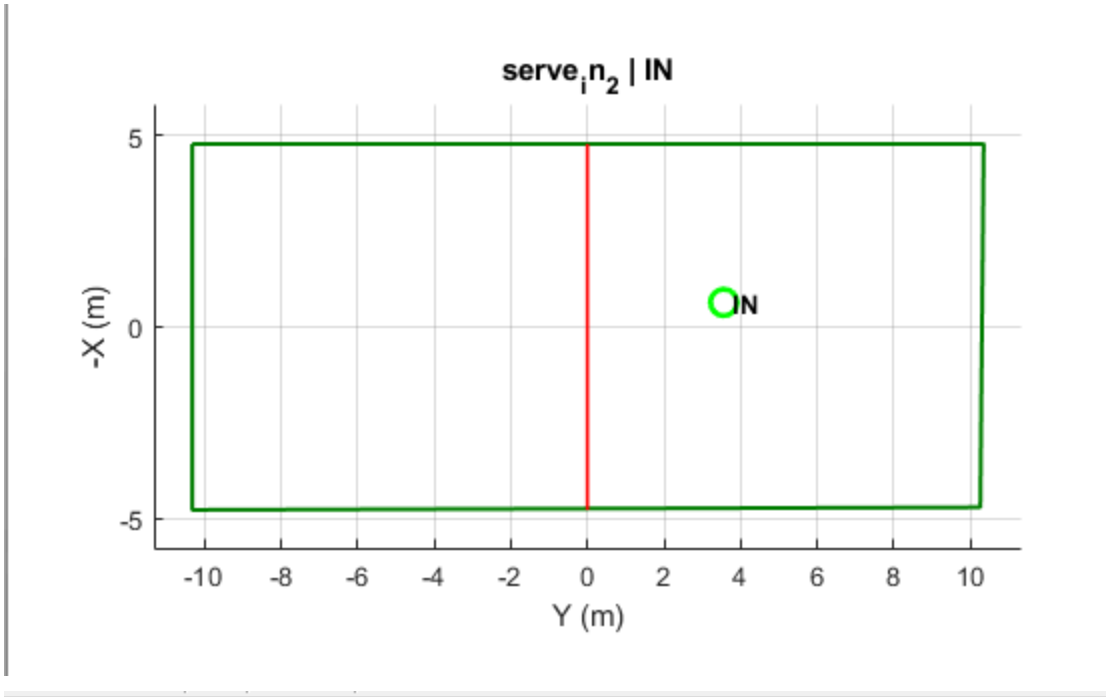

File Edit View Insert Tools Desktop Window Help
serve_net_1 | NET
NET
-X (m)
Y (m)
5
0
-5
-10
-8
-6
-4
-2
0
2
4
6
8
10

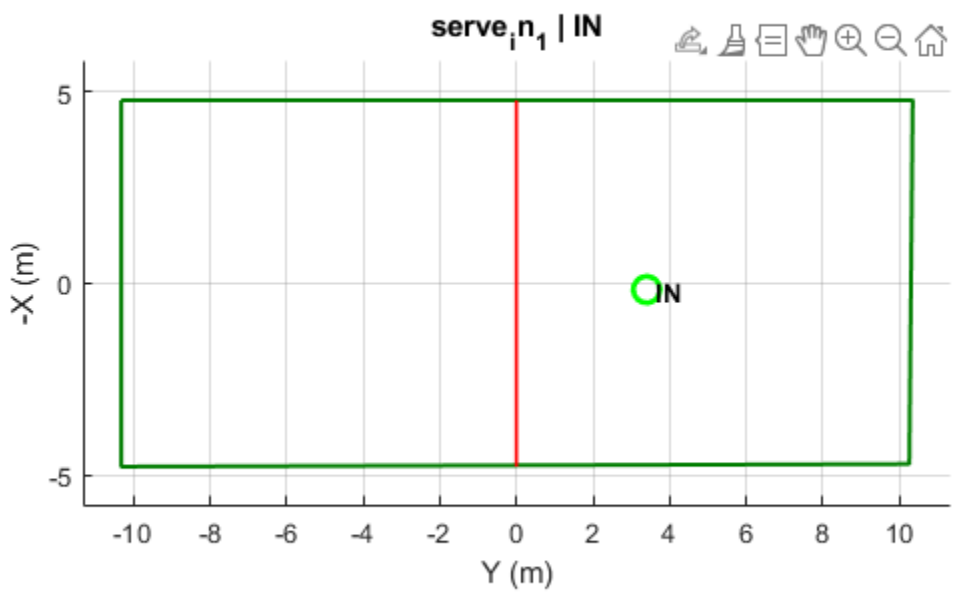


(outside of serve box)


serve_out_1 | OUT
5
0
-5
-X (m)
OUT
-10
-8
-6
-4
-2
0
2
4
6
8
10
Y (m)


serve_out_1 | OUT
0
-2
-4
-6
-X (m)
OUT
-6
-4
-2
0
2
4
6
8
10
Y (m)

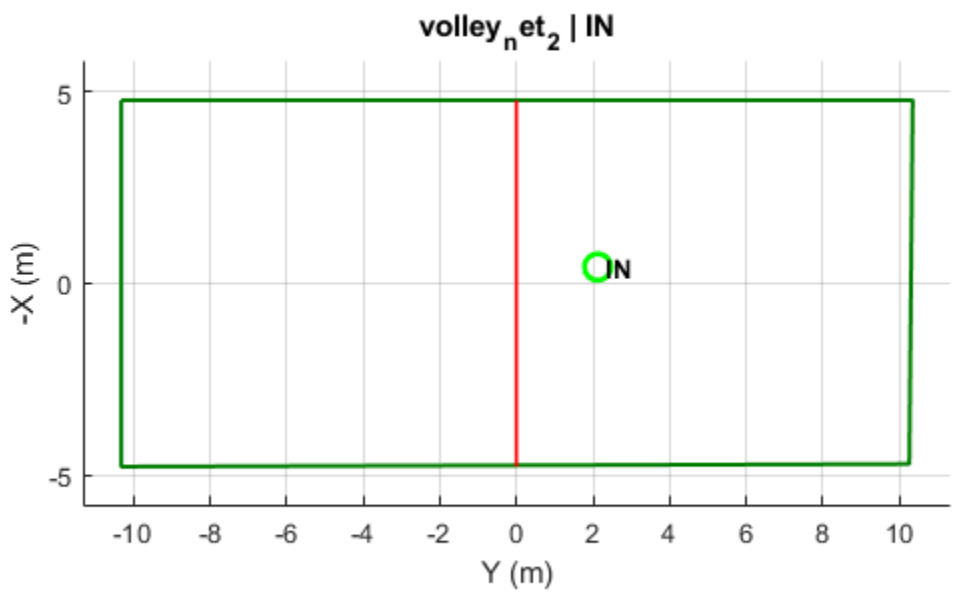


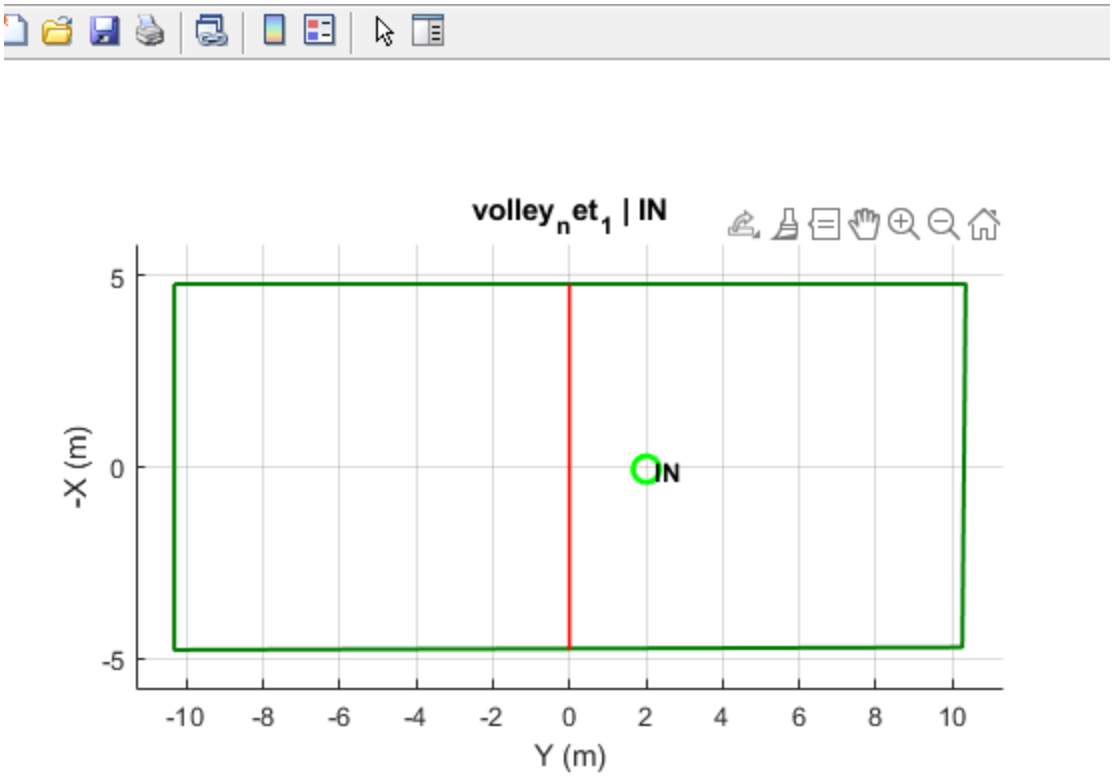


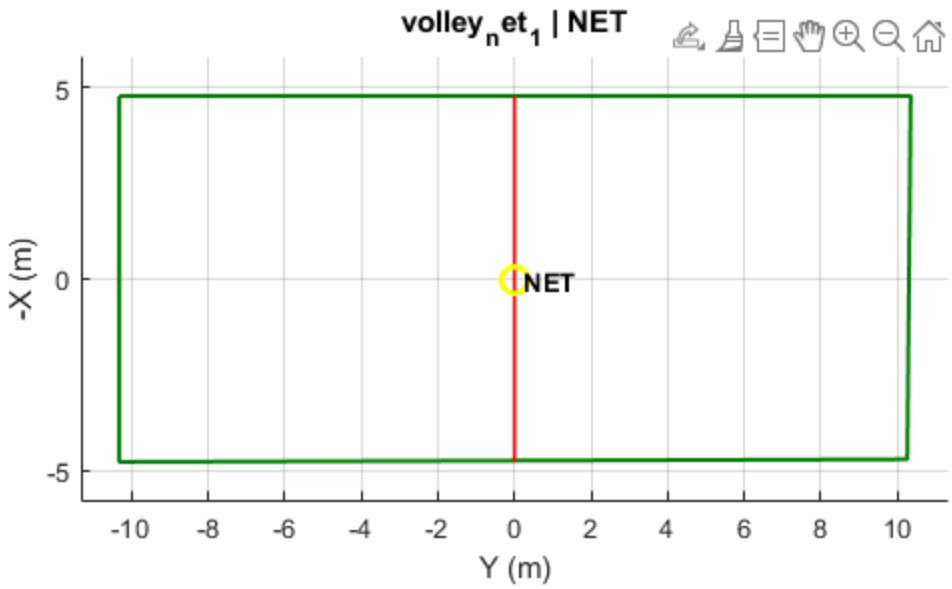
Preset serve_net_2 | bounceIdx = 18
Bounce XYZ = [0.2880 0.0160 0.3595]
=== SERVE RESULT ===

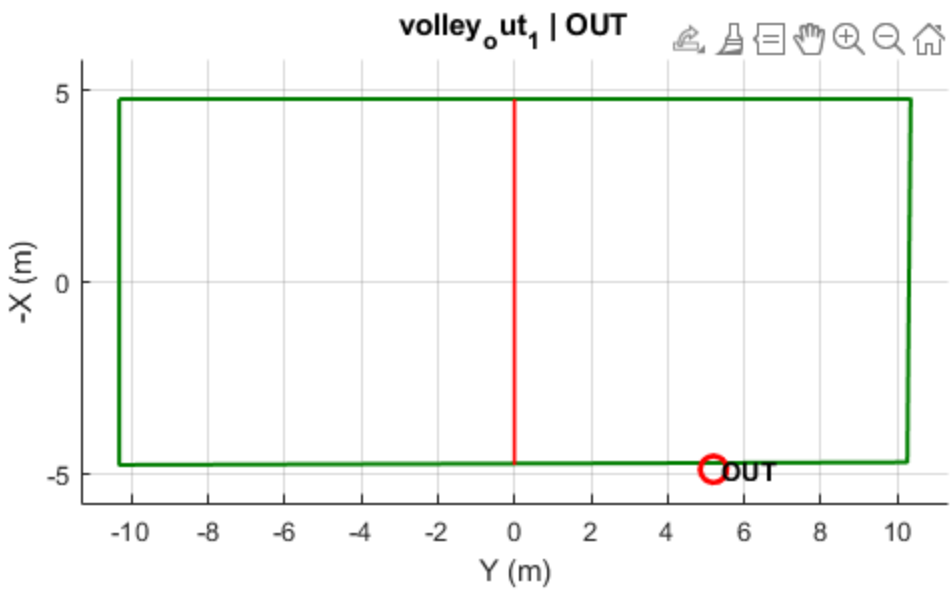
Preset	BounceX	BounceY	BounceZ	Decision	ValidFrames

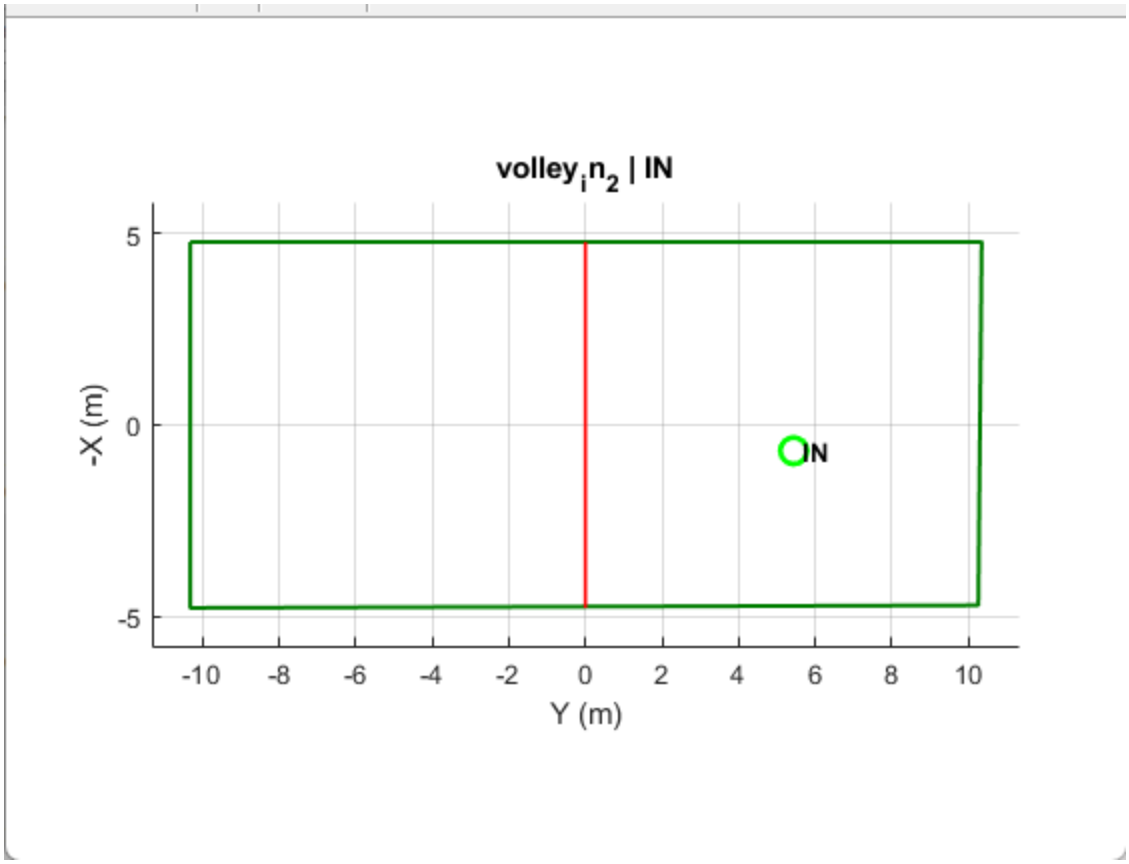
```
{'serve_in_1' }     0.16       3.4     0.35951    {'IN' }        21
{'serve_in_2' }    -0.64      3.56     0.35951    {'IN' }        22
{'serve_out_1'}     4.64       3.4     0.35951    {'OUT'}        21
{'serve_out_2'}    0.088      7.92     0.35951    {'OUT'}        22
{'serve_net_1'}        0      0.16     0.35951    {'NET'}        23
{'serve_net_2'}    0.288     0.016     0.35951    {'NET'}        24

>>
```



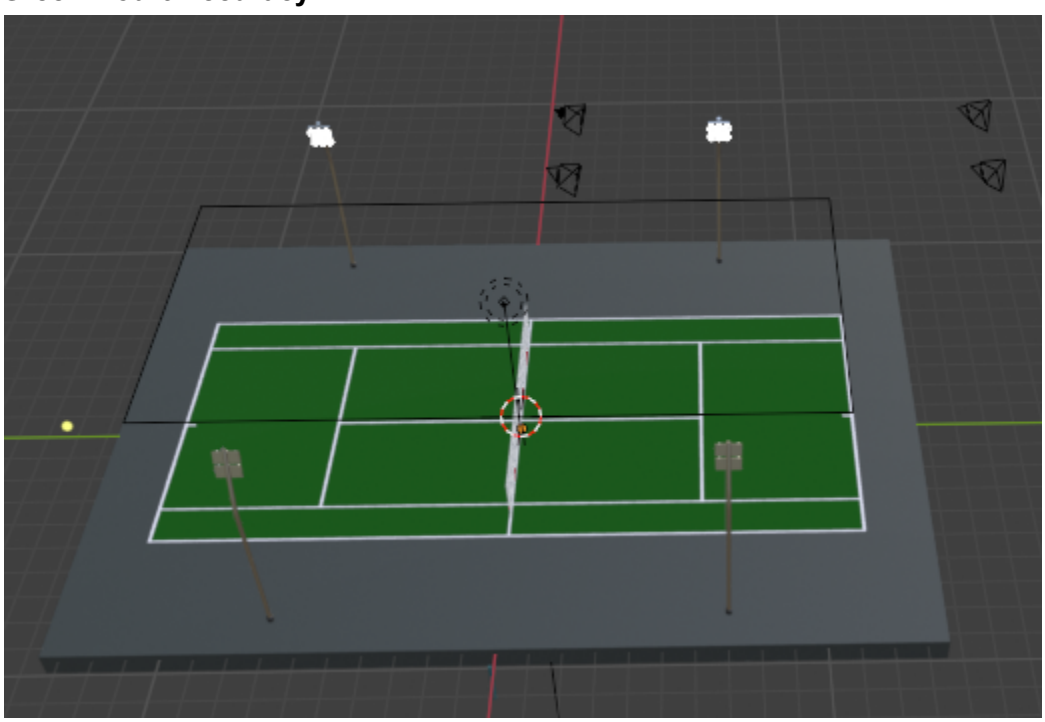

volley_net_1 | IN
5
0
-5
-X (m)
IN
-10
-8
-6
-4
-2
0
2
4
6
8
10
Y (m)

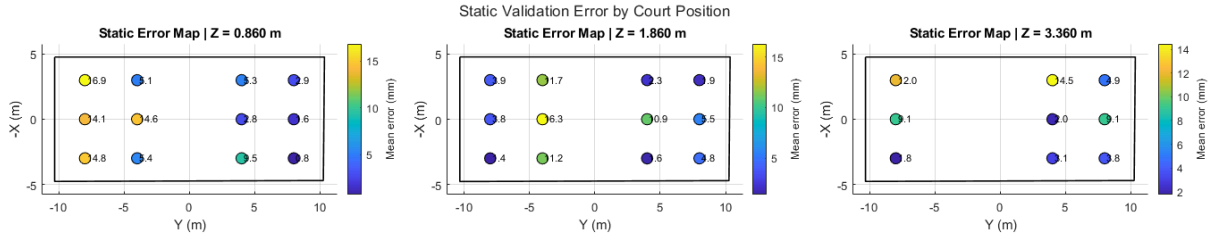

volley_net_1 | NET
NET
-X (m)
Y (m)
5
0
-5
-10
-8
-6
-4
-2
0
2
4
6
8
10

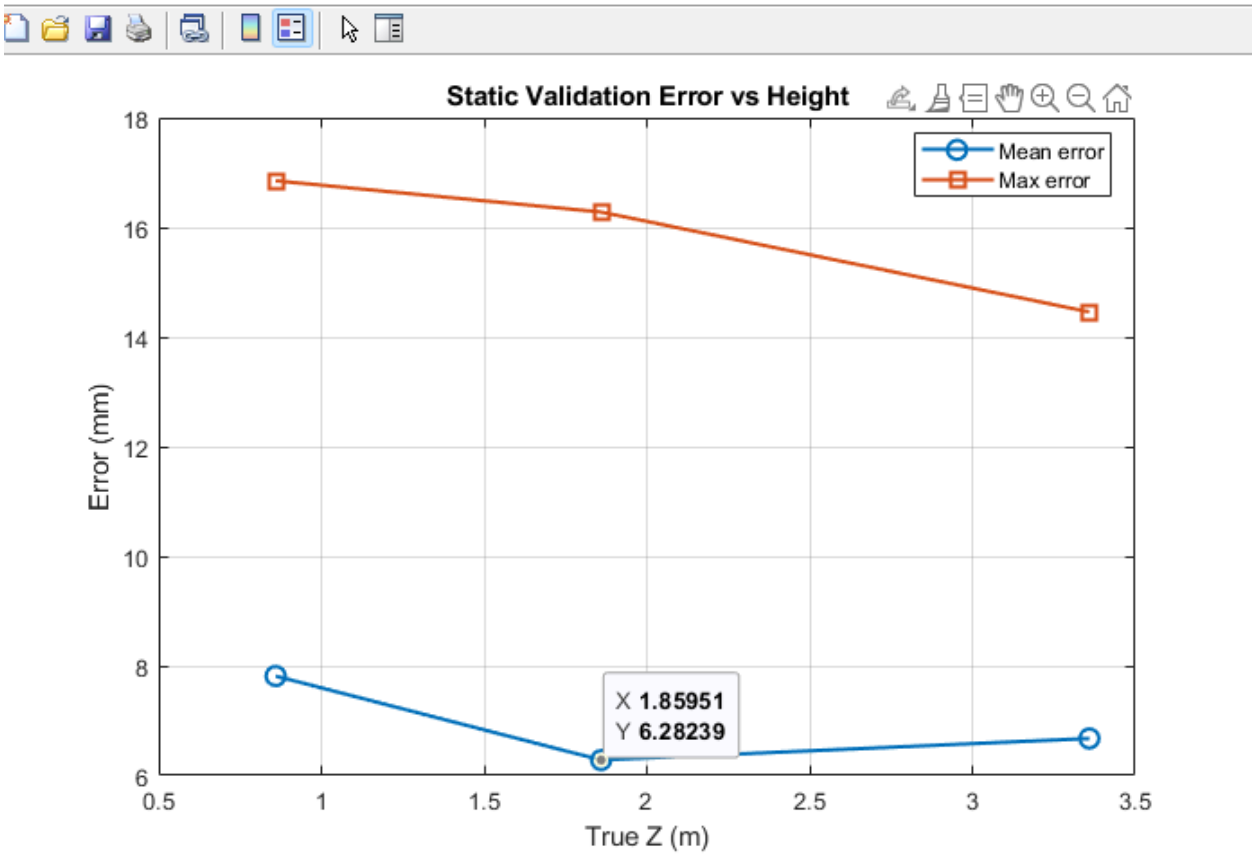

volley_out_1 | OUT
5
0
-5
-X (m)
OUT
-10
-8
-6
-4
-2
0
2
4
6
8
10
Y (m)



Green Court Accuracy:







=== SHOT VALIDATION ===

preset	type	n	meanDx	meanDy	meanDz	meanE	maxE	rmseE	bounceE
"serve_default"	"serve"	8	0.00038935	-0.0026541	0.0051649	0.0088385	0.025514	0.012117	0.012174
"serve_t"	"serve"	8	-0.00015222	0.0021847	-0.00044423	0.0044685	0.022265	0.0081775	0.0010745
"serve_wide"	"serve"	8	0.00093404	-0.00045547	0.00054769	0.0045005	0.014925	0.0063211	0.0025008
"serve_body"	"serve"	8	-2.1672e-05	-0.0031155	0.0055953	0.0078732	0.020241	0.010687	0.010083
"volley_default"	"volley"	6	0.00032515	-0.0030356	0.0033819	0.0055141	0.020789	0.0087864	0.0017866
"volley_short"	"volley"	7	-0.00013477	-0.0008976	0.0025036	0.0047747	0.0078066	0.0053034	0.0078066
"volley_deep"	"volley"	6	0.00027629	-0.0011956	0.00090234	0.0067482	0.019799	0.0096552	0.0022469
"volley_cross"	"volley"	8	0.0003904	-0.0035089	0.0079403	0.010316	0.027772	0.01283	0.0073156

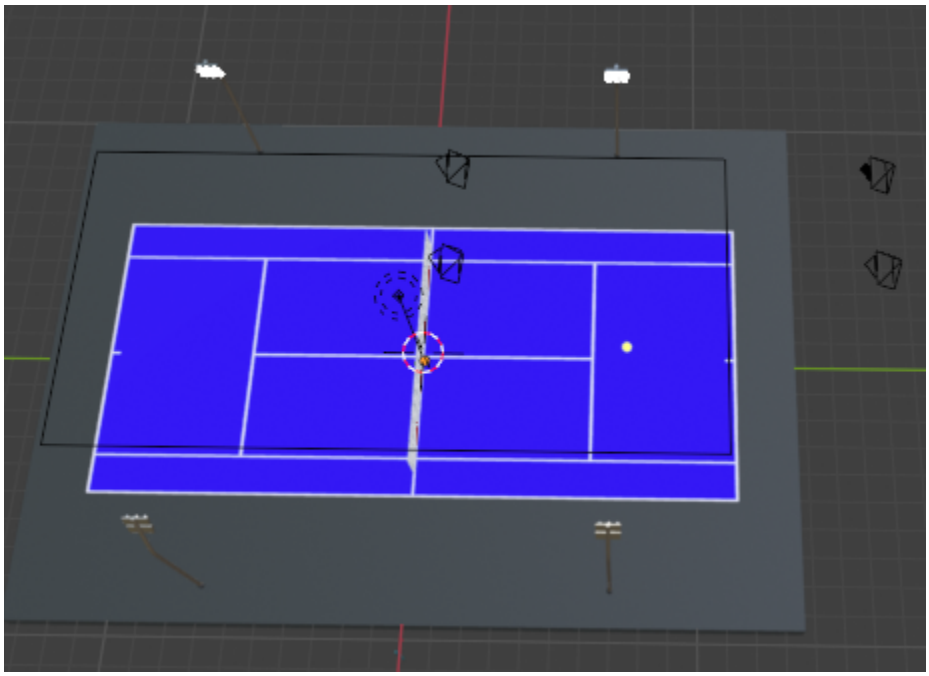
=== STATIC SUMMARY ===

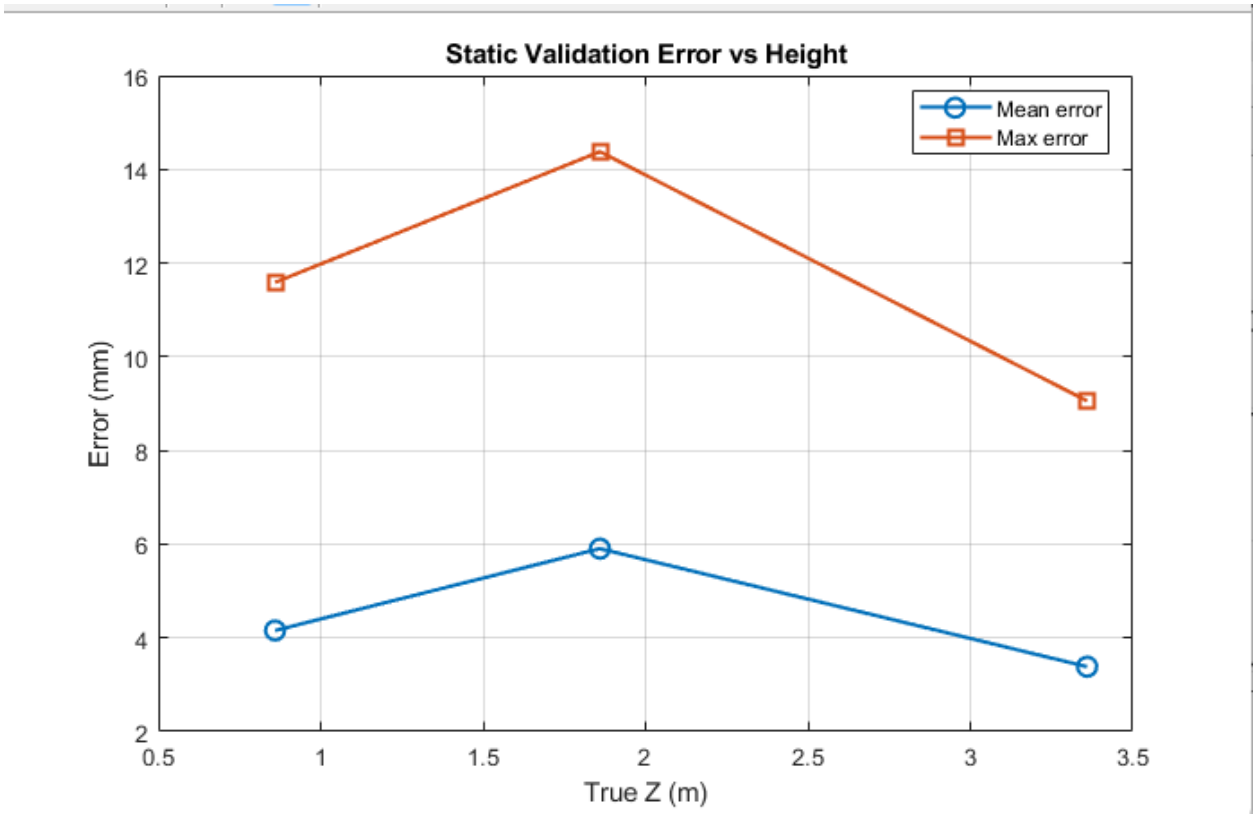
meanDx	meanDy	meanDz	meanE	maxE	rmseE	pair1Count	pair2Count
0.0002335	-0.0019642	0.0028008	0.006947	0.016851	0.0085716	45	54

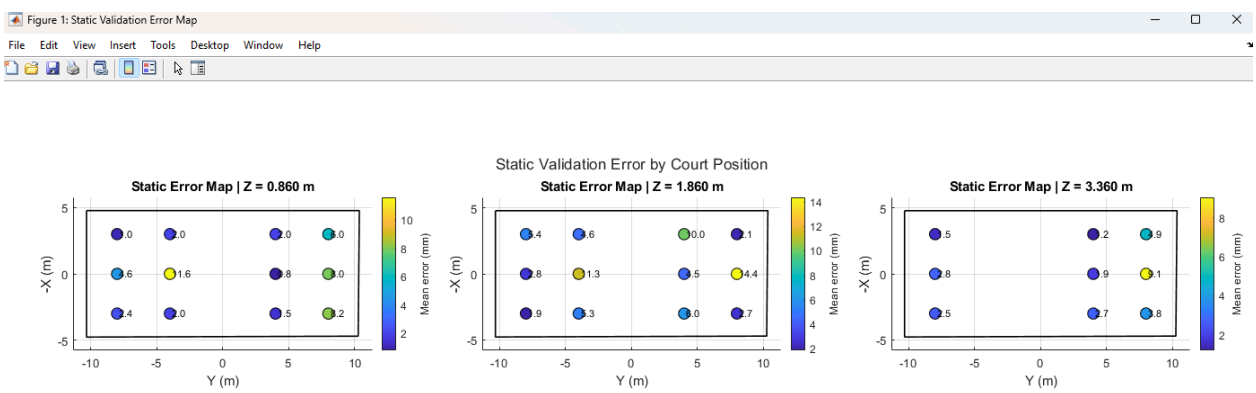
=== SHOT SUMMARY ===

serveMeanE	serveMaxE	serveBounce	volleyMeanE	volleyMaxE	volleyBounce
0.0064202	0.025514	0.006458	0.0068383	0.027772	0.0047889

Blue Court:








```
=== SHOT VALIDATION ===
```

preset	type	n	meanDx	meanDy	meanDz	meanE	maxE	rmseE	bounceE
"serve_default"	"serve"	8	-0.00021288	0.00047671	-0.00023458	0.0045515	0.0093141	0.0052325	0.0023802
"serve_t"	"serve"	8	9.255e-05	-0.0024665	0.0037465	0.0050458	0.0080776	0.0057291	0.0080776
"serve_wide"	"serve"	8	0.0010617	8.6015e-05	0.00040444	0.0045996	0.014925	0.0062399	0.0054493
"serve_body"	"serve"	8	5.8219e-05	0.00078319	-0.00071876	0.0035142	0.0062088	0.0039348	0.0041662
"volley_default"	"volley"	6	0.00039785	-0.0040818	0.0053353	0.0068165	0.012779	0.0080743	0.00060172
"volley_short"	"volley"	7	-1.8024e-05	3.5663e-05	-0.00074	0.00264	0.0036619	0.0027746	0.00091187
"volley_deep"	"volley"	6	0.0014812	-6.9649e-05	0.001152	0.0062902	0.014933	0.0077856	0.009146
"volley_cross"	"volley"	8	-0.00020778	0.00019637	0.0038481	0.0067931	0.027772	0.010627	0.0072091

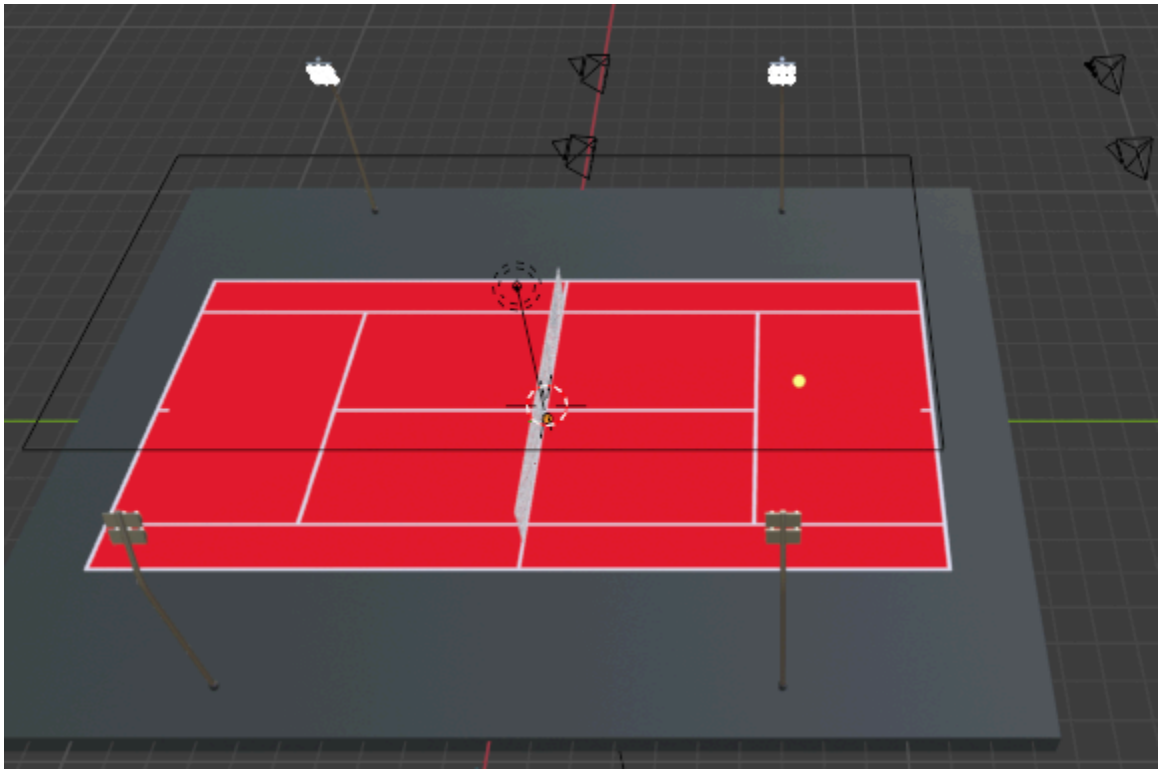
```
=== STATIC SUMMARY ===
```

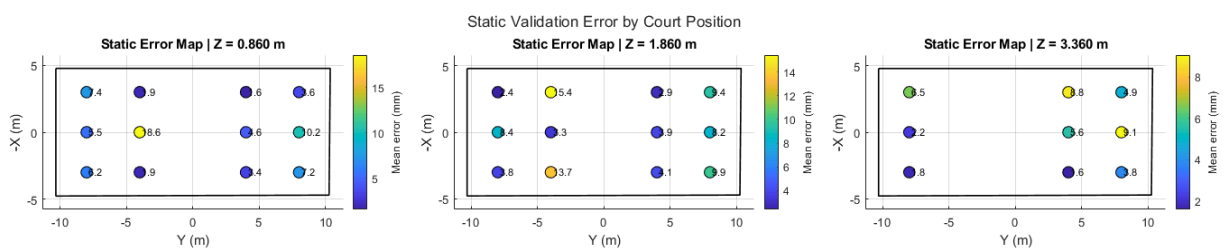
meanDx	meanDy	meanDz	meanE	maxE	rmseE	pair1Count	pair2Count
7.0254e-06	-0.0012259	0.0020198	0.0045819	0.014377	0.0057281	45	54

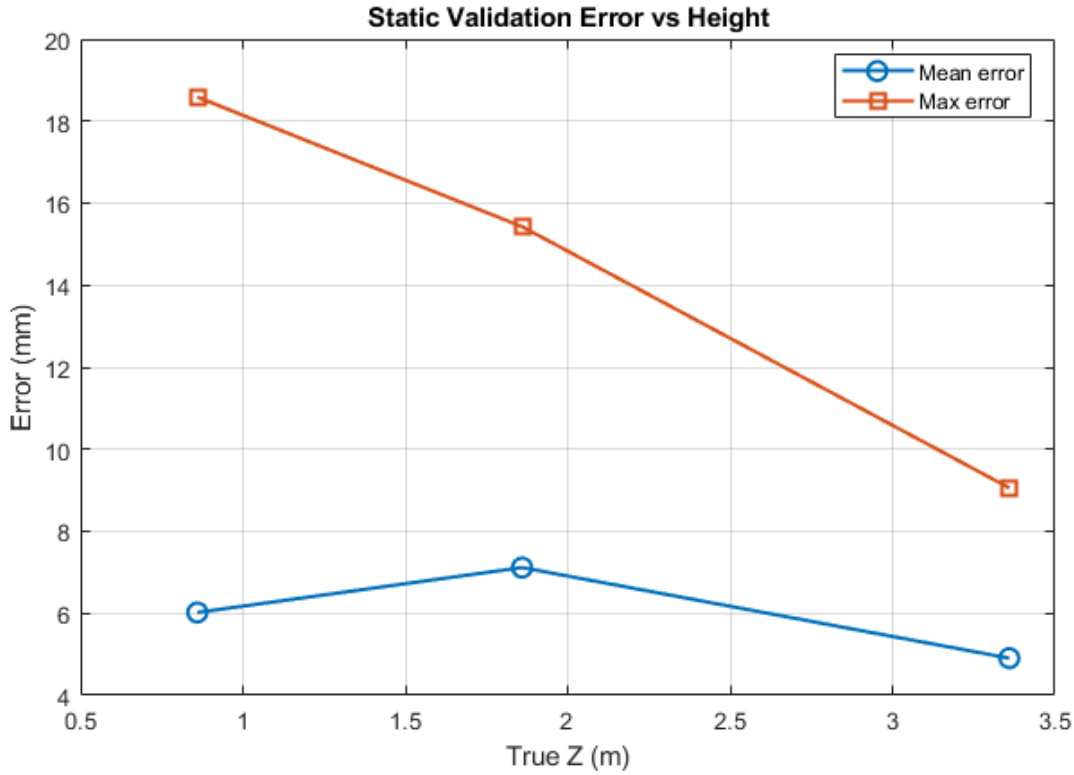
```
=== SHOT SUMMARY ===
```

serveMeanE	serveMaxE	serveBounce	volleyMeanE	volleyMaxE	volleyBounce
0.0044278	0.014925	0.0050184	0.005635	0.027772	0.0044672

RedCourt:








```
=== SHOT VALIDATION ===
```

preset	type	n	meanDx	meanDy	meanDz	meanE	maxE	rmseE	bounceE
"serve_default"	"serve"	8	9.1858e-05	-0.0025908	0.0041643	0.0066066	0.015609	0.007977	0.0063299
"serve_t"	"serve"	8	-0.00027921	-0.00061446	0.00097651	0.0038506	0.0076605	0.0046776	0.0057318
"serve_wide"	"serve"	8	0.0012625	0.00099443	-0.00097607	0.0036914	0.014925	0.0057355	0.00042364
"serve_body"	"serve"	8	-6.9825e-05	-0.001582	0.0026236	0.0044881	0.0099664	0.0055154	0.0031833
"volley_default"	"volley"	6	-0.00039782	0.00062601	-0.00067869	0.0052578	0.014554	0.0072511	0.0011641
"volley_short"	"volley"	7	-0.0001575	-0.0023694	0.0038879	0.0064639	0.014905	0.0074524	0.0039531
"volley_deep"	"volley"	6	0.00071135	-0.00044067	0.00090916	0.0043698	0.0063986	0.0047648	0.0063986
"volley_cross"	"volley"	8	-0.0001454	-0.0012272	0.0054911	0.0069923	0.027772	0.010844	0.0072091

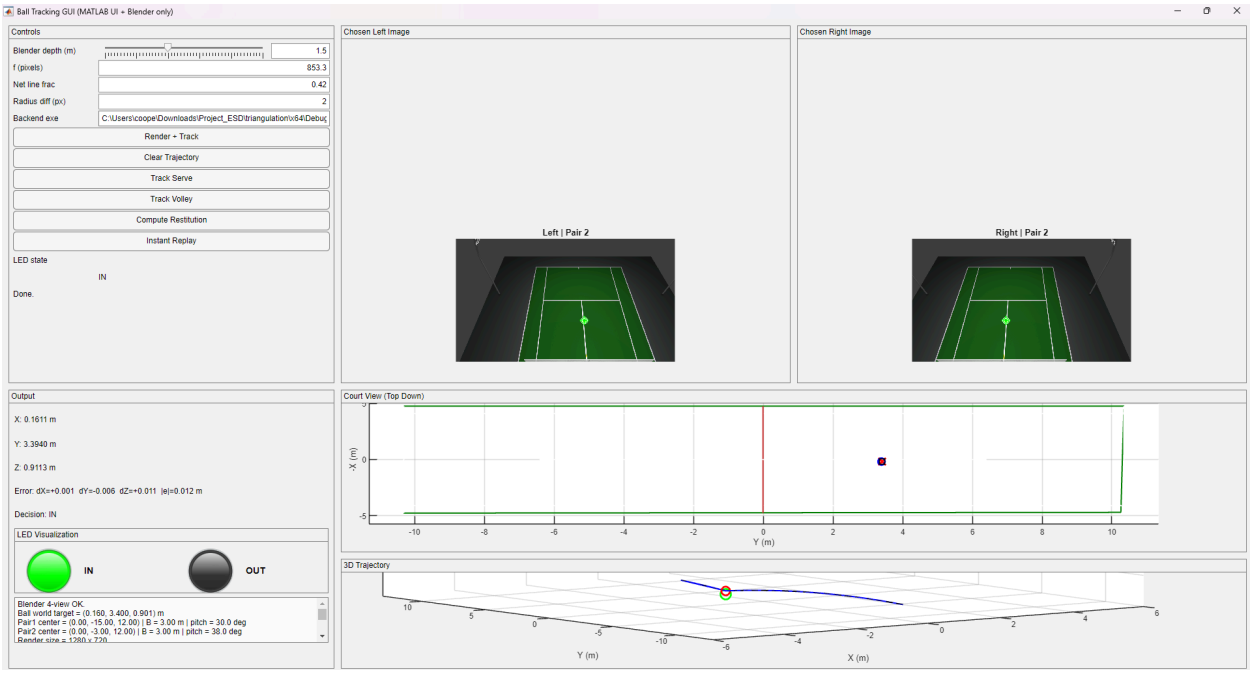
```
=== STATIC SUMMARY ===
```

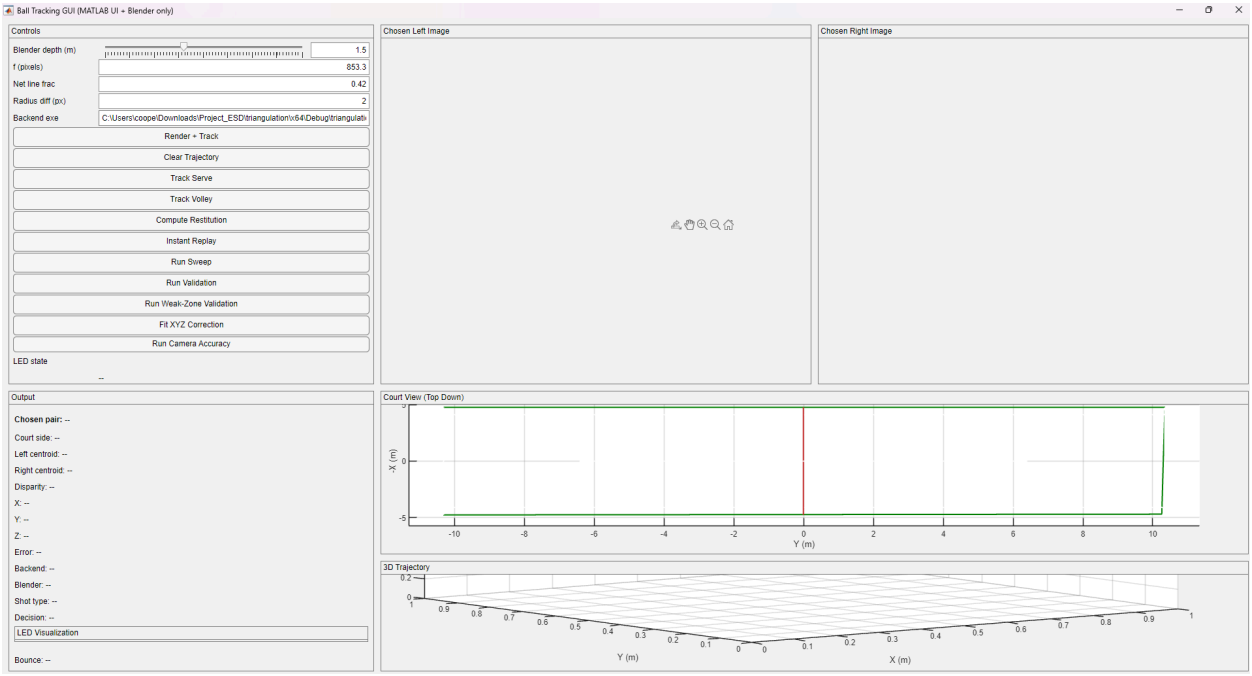
meanDx	meanDy	meanDz	meanE	maxE	rmseE	pair1Count	pair2Count
6.9343e-05	-0.0022395	0.0024633	0.0061149	0.018589	0.0073538	45	54

```
=== SHOT SUMMARY ===
```

serveMeanE	serveMaxE	serveBounce	volleyMeanE	volleyMaxE	volleyBounce
0.0046592	0.015609	0.0039172	0.0057709	0.027772	0.0046812

Professional GUI (customer use) versus developmental GUI (for more stats/accuracy options):


Ball Tracking GUI (MATLAB UI + Blender only)
Controls
Blender depth (m)
1.5
f (pixels)
853.3
Net line frac
0.42
Radius diff (px)
2
Backend exe
C:\Users\coope\Downloads\Project_ESD\triangulation\x64\Debug
Render + Track
Clear Trajectory
Track Serve
Track Volley
Compute Restitution
Instant Replay
LED state
IN
Done.
Chosen Left Image
Left | Pair 2
Chosen Right Image
Right | Pair 2
Output
X: 0.1611 m
Y: 3.3940 m
Z: 0.9113 m
Error: dX=+0.001 dY=-0.006 dZ=+0.011 |e|=0.012 m
Decision: IN
LED Visualization
IN
OUT
Blender 4-view OK.
Ball world target = (0.160, 3.400, 0.901) m
Pair1 center = (0.00, -15.00, 12.00) | B = 3.00 m | pitch = 30.0 deg
Pair2 center = (0.00, -3.00, 12.00) | B = 3.00 m | pitch = 38.0 deg
Render size = 1280 x 720
Court View (Top Down)
X (m)
Y (m)
3D Trajectory
Y (m)
X (m)


Ball Tracking GUI (MATLAB UI + Blender only)
Controls
Blender depth (m)
1.5
f (pixels)
853.3
Net line frac
0.42
Radius diff (px)
2
Backend exe
C:\Users\coope\Downloads\Project_ESD\triangulation\x64\Debug\triangulati
Render + Track
Clear Trajectory
Track Serve
Track Volley
Compute Restitution
Instant Replay
Run Sweep
Run Validation
Run Weak-Zone Validation
Fit XYZ Correction
Run Camera Accuracy
LED state
--
Chosen Left Image
Chosen Right Image
Output
Chosen pair: --
Court side: --
Left centroid: --
Right centroid: --
Disparity: --
X: --
Y: --
Z: --
Error: --
Backend: --
Blender: --
Shot type: --
Decision: --
LED Visualization
Bounce: --
Court View (Top Down)
X (m)
Y (m)
3D Trajectory
Y (m)
X (m)